

CS19541 – COMPUTER NETWORKS LAB

Practical 10 – Internetworking in Cisco Packet Tracer

AIM

To configure internetworking using routers and a wireless router in Cisco Packet Tracer and verify communication between devices.

REQUIREMENTS

Cisco Packet Tracer, routers, PCs, laptop, wireless router, cable modem, Internet cloud, and server.

PART A – ROUTER-BASED INTERNETWORKING

Configure Router1 with two networks. Set FastEthernet0/0 to 192.168.10.1/24 and FastEthernet0/1 to 192.168.20.1/24. Configure PC0 as 192.168.10.2 with gateway 192.168.10.1, and PC1 as 192.168.20.2 with gateway 192.168.20.1.

ROUTER COMMANDS

```
enable
configure terminal
interface fastethernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface fastethernet0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
```

PART B – WIRELESS ROUTER AND DHCP

Set the wireless network name (SSID) to HomeNetwork and enable DHCP. Configure DNS as 208.67.220.220. Connect the laptop wirelessly and set the PC to obtain its IP address automatically. Configure the server's DHCP pool with starting IP 208.67.220.1, subnet mask 255.255.255.0, DNS 208.67.220.220, and a maximum of 50 users.

PROCEDURE

1. Place the required devices in the workspace.
2. Connect PCs to the router using straight-through cables.
3. Configure router interfaces and PC IP settings.
4. Build the wireless topology and configure SSID and DHCP.
5. Connect the laptop to the wireless network.
6. Use Simple PDU or ping to test connectivity.

INPUT

IP addresses, subnet masks, default gateways, wireless SSID, and DHCP settings.

OUTPUT

The devices receive the configured IP settings and communicate across the connected networks. Connectivity can be verified using ping or Simple PDU.

OBSERVATION

Router interfaces were configured and enabled. PCs were assigned IP addresses and default gateways. Wireless access and DHCP settings were configured, and connectivity was tested.

RESULT

Internetworking using routers and a wireless router was configured and verified successfully in Cisco Packet Tracer.